

Math 526, F16
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score	
1	20		
2	20		
3	26		
4	20		
5	18		
Total	104		

Good Luck !

Problem 1. (20 points) Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ -4 \\ 4 \end{bmatrix}$.

(a) Reduce $[A \ b]$ to $[R \ d]$ where R denotes the reduced echelon form of A .

$$\begin{aligned}
 [A|b] &= \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 2 & 4 & 8 & 10 & -4 \\ 3 & 6 & 7 & 10 & 4 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & -4 \\ 0 & 0 & -2 & -2 & 4 \end{array} \right] \\
 &\rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \\
 &\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 6 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] = [R|d]
 \end{aligned}$$

(b) Find a particular solution x_p of $Ax = b$ by setting the free variables to be zero.

$$\begin{aligned}
 Ax_p = b &\Leftrightarrow Rx_p = d \rightarrow \begin{cases} x_1 + 2x_2 + x_4 = 6 \\ x_3 + x_4 = -2 \end{cases} \\
 \text{free variables: } x_2, x_4 &\rightarrow \text{setting } x_2 = x_4 = 0 \\
 \Rightarrow \begin{cases} x_1 = 6 \\ x_3 = -2 \end{cases} &\Rightarrow x_p = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}
 \end{aligned}$$

(c) Find the complete solution to $Ax = b$.

$$\begin{aligned}
 \text{Two special solutions: } Ax_n = 0 &\Leftrightarrow Rx_n = 0 \\
 \begin{cases} x_1 + 2x_2 + x_4 = 0 \\ x_3 + x_4 = 0 \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 \text{i) Set } x_2 = 1, x_4 = 0, &\text{ we get } x_1 = -2, x_3 = 0 \\
 \Rightarrow S_1 &= (-2, 1, 0, 0)
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) Set } x_2 = 0, x_4 = 1, &\text{ we get } x_1 = -1, x_3 = -1 \\
 \Rightarrow S_2 &= (-1, 0, -1, 1)
 \end{aligned}$$

$$\text{Thus } x = x_p + x_n = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix} + C_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

Problem 2. (20 points)

Let $A = \begin{bmatrix} 2 & 4 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) Find the dimension and a basis for the row space of A , $C(A^T)$.

$$\dim(C(A^T)) = 2 \leftarrow \text{Pivot variables: } x_1, x_3$$

$$\text{a basis: } \left\{ \begin{bmatrix} 2 \\ 4 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim(C(A)) = 2$$

$$\text{a basis: } \left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

(c) Find the dimension and a basis for the nullspace of A , $N(A)$.

$$\text{Free variables: } x_2, x_4 \Rightarrow \dim(N(A)) = 2$$

$$\text{Solve } AX = 0, \text{ i.e. } \begin{cases} 2x_1 + 4x_2 + 5x_4 = 0 \\ x_3 + 2x_4 = 0 \end{cases}$$

$$\text{i) set } x_2 = 1, x_4 = 0, \text{ we get } x_1 = -2, x_3 = 0 \\ \Rightarrow s_1 = (-2, 1, 0, 0)$$

$$\text{ii) set } x_2 = 0, x_4 = 1, \text{ we get } x_1 = -\frac{5}{2}, x_3 = -2 \\ \Rightarrow s_2 = (-\frac{5}{2}, 0, -2, 1)$$

$$\Rightarrow \text{a basis of } N(A) : \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5/2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

(d) Find the dimension and a basis for the left nullspace of A , $N(A^T)$.

$$N(A^T) = 3 - \dim(C(A^T)) = 3 - 2 = 1$$

$$\text{Solve } A^T y = 0 \Rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 4 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 2 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{cases} 2y_1 = 0 \\ 4y_2 = 0 \\ y_2 = 0 \\ 5y_1 + 2y_2 = 0 \end{cases} \Rightarrow y_1 = y_2 = 0 \Rightarrow y = \begin{bmatrix} 0 \\ 0 \\ y_3 \end{bmatrix} = y_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ \Rightarrow \text{a basis of } N(A^T) = \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Problem 3. (26 points)

(a) Find the straight line $C + Dt$ that comes closest to the values $\mathbf{b} = 0, 8, 8, 20$ at times $t = 0, 1, 3, 4$ by the least squares fitting.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

solve $A^T A x = A^T b$

$$\begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$\Rightarrow C=1, D=4 \Rightarrow$ the least squares fitting line is $y = 1 + 4t$

(b) Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{bmatrix}$. Find an orthonormal basis $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ for the column space of A using the Gram-Schmidt process.

$$\tilde{\mathbf{q}}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \|\tilde{\mathbf{q}}_1\|^2 = 1^2 + 1^2 + 2^2 = 6$$

$$\tilde{\mathbf{q}}_2 = \mathbf{a}_2 - \frac{\tilde{\mathbf{q}}_1^T \mathbf{a}_2}{\|\tilde{\mathbf{q}}_1\|^2} \tilde{\mathbf{q}}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \frac{\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \frac{0}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\|\tilde{\mathbf{q}}_2\|^2 = 1^2 + (-1)^2 = 2$$

$$\tilde{\mathbf{q}}_3 = \mathbf{a}_3 - \frac{\tilde{\mathbf{q}}_1^T \mathbf{a}_3}{\|\tilde{\mathbf{q}}_1\|^2} \tilde{\mathbf{q}}_1 - \frac{\tilde{\mathbf{q}}_2^T \mathbf{a}_3}{\|\tilde{\mathbf{q}}_2\|^2} \tilde{\mathbf{q}}_2$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \frac{\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \frac{9}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad \|\tilde{\mathbf{q}}_3\|^2 = (-1)^2 + (-1)^2 + 1^2 = 3$$

$$\Rightarrow \mathbf{q}_1 = \frac{\tilde{\mathbf{q}}_1}{\|\tilde{\mathbf{q}}_1\|} = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}, \quad \mathbf{q}_2 = \frac{\tilde{\mathbf{q}}_2}{\|\tilde{\mathbf{q}}_2\|} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}, \quad \mathbf{q}_3 = \frac{\tilde{\mathbf{q}}_3}{\|\tilde{\mathbf{q}}_3\|} = \begin{bmatrix} -1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Problem 4. (20 points) Let $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$.

(a) Solve the linear system $Ax = b$ with $b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ by Cramer's Rule, $x_i = \frac{\det(B_i)}{\det(A)}$.

$$\det A = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = -1 \cdot \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = (-1)(-1) + 3 \cdot 1 = 4.$$

$$\det B_1 = \begin{vmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1 \Rightarrow x_1 = \frac{\det B_1}{\det A} = -\frac{1}{4}$$

$$\det B_2 = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix} = - \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1 \Rightarrow x_2 = \frac{\det B_2}{\det A} = \frac{1}{4}$$

$$\det B_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1 \Rightarrow x_3 = \frac{1}{4}$$

(b) Find A^{-1} using the cofactor formula, $A^{-1} = \frac{C^T}{\det A}$.

$$C = \begin{bmatrix} -1 & 1 & 1 \\ 2 & -6 & 2 \\ 1 & 3 & -1 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow A^{-1} &= \frac{C^T}{\det A} = \frac{1}{4} \begin{bmatrix} -1 & 2 & 1 \\ 1 & -6 & 3 \\ 1 & 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} -1/4 & 1/2 & 1/4 \\ 1/4 & -3/2 & 3/4 \\ 1/4 & 1/2 & -1/4 \end{bmatrix} \end{aligned}$$

Problem 5. (18 points) True or False (no explanation needed).

(a) If the columns of a matrix are independent, so are the rows.

False

(b) If $\mathbf{A}^T = -\mathbf{A}$ then the row space of $\mathbf{A}$ equals the column space of $\mathbf{A}$.

True

(c) The nullspace $N(\mathbf{A})$ and the row space $C(\mathbf{A}^T)$ are orthogonal complements.

True

(d) If $\mathbf{P}$ is a projection matrix, then $\mathbf{P}^2 = \mathbf{P}$.

True

$$(e) \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} -d & -e & f \\ -a & -b & c \\ -g & -h & i \end{vmatrix}.$$

False

(f) The area of the triangle formed by the three points $(-1, 2)$, $(0, 5)$, $(3, 3)$ is $\frac{1}{2}\det\mathbf{A}$ with

$$\mathbf{A} = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 5 & 1 \\ 3 & 3 & 1 \end{bmatrix}.$$

False